

Electric Fields and Van Der Waals Forces

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Abstract

This report investigates the electric field and potential of a physical dipole and connects those results to the real-world context of Van der Waals intermolecular forces. Beginning with hand-drawn and computer visualizations of the dipole field and equipotential lines, we derive analytic expressions for the electric potential and field both along the dipole axis and at a general point in the plane, using the principle of superposition and Taylor series approximations. We then apply these results to the water molecule, corroborating the Wikipedia value of its dipole moment (1.8546 D) through both experimental literature review and a geometric calculation based on VSEPR geometry. Using a dipole model of water, we estimate the electric field between neighboring water molecules and the electric potential energy for several dipole-pair orientations. We also derive the dipole–dipole force and its generalization to a polarizable molecule. Finally, we argue that the rapid fall-off of multipole fields — from $1/r^2$ for a monopole to $1/r^3$ for a dipole and $1/r^4$ for a quadrupole — explains why macroscopic objects appear to have sharp, solid boundaries.

Hand-Drawn Visualizations

Figure ?? shows our hand-drawn visualization of the electric field lines (solid black curves) and equipotential surfaces (dashed gold curves) for a physical dipole consisting of charges $+q$ and $-q$.

Electric Potential and Field Along the Dipole Axis

Setup

Consider a physical dipole with charge separation d , centered at the origin, with $+q$ at $x = +\frac{d}{2}$ and $-q$ at $x = -\frac{d}{2}$, as shown in Figure ??.

(a) Electric potential on the axis

Using $V = kq/r$ and the principle of superposition, the potential at a point x on the x -axis (with $x > \frac{d}{2}$) is:

$$V = \frac{k(-q)}{\frac{1}{2}d + x} + \frac{k(q)}{x - \frac{1}{2}d}$$

(b) Electric field on the axis

Since $E = F_{\text{sources on probe}}/q_{\text{probe}}$ and $F = kq_1q_2/r^2$, with only an x -component, the x -component of the field is:

$$E_x = \frac{k(-q)}{\left(\frac{1}{2}d + x\right)^2} \hat{i} + \frac{k(q)}{\left(x - \frac{1}{2}d\right)^2} \hat{i}$$

(c) Simplified forms using the dipole moment

Combining over a common denominator with $p = qd$:

$$V = \frac{kp}{x^2 - \frac{1}{4}d^2}$$

Taking the limit $x \gg d$:

$$V \approx \frac{kp}{x^2}$$

The electric field follows from $E_x = -dV/dx$:

$$E_x = \frac{2kp}{x^3}$$

General Expression for the Electric Field of a Physical Dipole

(a) Distances from charges to a generic point

For a generic point (x, y) with $r^2 = x^2 + y^2$, the distances from each charge are:

$$r_{q+} = \sqrt{(x - \frac{d}{2})^2 + y^2}, \quad r_{q-} = \sqrt{(x + \frac{d}{2})^2 + y^2}$$

Applying superposition, the electric potential is:

$$V = \frac{k(-q)}{r_{q-}} + \frac{k(q)}{r_{q+}} = \frac{-kq}{\sqrt{(x + \frac{d}{2})^2 + y^2}} + \frac{kq}{\sqrt{(x - \frac{d}{2})^2 + y^2}}$$

(b) Electric field components

Taking $\vec{E} = -\nabla V$ (partial derivatives computed with a computer algebra system):

$$E_x = -\frac{\partial V}{\partial x} = \frac{-kq(2x + 0.5d)}{(y^2 + (x+0.5d)^2)^{3/2}} + \frac{kq(x - 0.5d)}{(y^2 + (x-0.5d)^2)^{3/2}}$$

$$E_y = -\frac{\partial V}{\partial y} = \frac{-kq y}{(y^2 + (x+0.5d)^2)^{3/2}} + \frac{kq y}{(y^2 + (x-0.5d)^2)^{3/2}}$$

The handwritten derivations for parts (a), (b), and (c) of this section are shown in Figures ??–?? below.

(c) Taylor series: potential falls off as $1/r^2$

We want to show that $V \propto 1/r^2$ in the limit $r \gg d$. Expanding $r_{q-} = \sqrt{(x + \frac{d}{2})^2 + y^2}$ using a Taylor series about $d = 0$:

$$r_{q-} \approx r + \frac{x}{2r} d \implies r_{q-} \approx r + \frac{1}{2} \frac{x}{r} d$$

Similarly:

$$r_{q+} \approx r - \frac{1}{2} \frac{x}{r} d$$

Substituting back and combining:

$$V \Rightarrow \frac{-kq r_{q+} + kq r_{q-}}{r_{q-} r_{q+}} \approx \frac{kq \frac{x}{r} d}{r^2}$$

Applying the approximations $x \approx r$ (so $x/r \approx 1$), $r = \sqrt{x^2 + y^2} \gg d$, and $p = qd$:

$$\boxed{V \approx \frac{kp}{r^2}}$$

The electric field magnitude then follows from $|\vec{E}| = -\partial V / \partial r$:

$$|\vec{E}| = -\frac{\partial}{\partial r} \left(\frac{kp}{r^2} \right) = \frac{2kp}{r^3}$$

showing that $|\vec{E}|$ falls off as $1/r^3$ in the limit $r \gg d$.

Computer Visualization of the Electric Field and Potential

Electric potential

Figure ?? shows a contour plot of the electric potential produced by Wolfram Mathematica using the expression derived in Section 3. The code used `ContourPlot` with $kq = 1$ and $d = 1$, and the electric potential expressions from question 3.

The Mathematica code used was:

```

kq = 1; d = 1;
ContourPlot[(-kq)/Sqrt[(x+0.5d)^2+y^2]+(kq)/Sqrt[(x-0.5d)^2+y^2],
{x,-5,5},{y,-5,5},PlotLegends->Automatic]

```

Electric field

Figure ?? shows the electric field vector plot. Code structure was provided by Professor Gresham and adapted to the x - and y -components derived in Section 3 with $kq = 1$ and $d = 1$.

The Mathematica code used was:

```

VectorPlot[{(-kq(x+0.5d))/(((y^2)+(x+0.5d)^2)^(3/2))
+(kq(x-0.5d))/(((y^2)+(x-0.5d)^2)^(3/2)),
(-kq y)/(((y^2)+(x+0.5d)^2)^(3/2))+((kq)y)/(((y^2)+(x-0.5d)^2)^(3/2))},
{x,-5,5},{y,-5,5},PlotLegends->Automatic]

```

Developing a Quantitative Sense of Fields and Forces Near Water Molecules

5.1. Corroboration of the Wikipedia Value of the Dipole Moment of Water

The Wikipedia article on Properties of Water reports a dipole moment of 1.8546 D for a water molecule. As Professor Gresham noted, this value is suspicious given its unusually high number of significant figures for such a small object. We used two approaches to assess its accuracy.

1. *Experimental approach*

After consulting the scientific literature, it turns out that the dipole moment of water changes depending on its physical state. Frank J. Lovas reports 1.855 D for gaseous water; Coulson and Eisenberg report 2.6 D for ice; Batista et al. found the average dipole moment could reach 3.09 D (Tu and Laaksonen). The Wikipedia value likely corresponds to gaseous water. The small discrepancy from Lovas's value can be attributed to rounding. A more cautious report

by Guggemos, Slavíček, and Kresin gives 1.85 D for the water monomer in gas phase and 1.9 D for the water tetramer. Notably, in 1973, Clough et al. directly measured the equilibrium dipole moment using the Stark effect, obtaining 1.8473 ± 0.0010 D — demonstrating that five-significant-figure precision is experimentally achievable.

2. Mathematical approach

Using the general dipole moment formula

$$\vec{\mu} = \sum_i q_i \vec{r}_i$$

(where q_i is the partial charge on atom i and $\vec{r}_i$ is its position from the origin), we placed the origin at the oxygen atom. The water molecule geometry according to VSEPR theory is shown in Figure ??.

The geometry data used:

- HOH angle: $\theta = 104.52$ (Huš and Urbic)
- O–H bond length: $r = 0.96$ Å (Siskos et al.)
- Dipole moment of O–H bond: $\mu_{\text{O-H}} = 1.5$ D (common textbook value)

The handwritten calculation of the vector sum is shown in Figure ??.

Since $\vec{r}_{1x} = -\vec{r}_{2x}$ and $\vec{r}_{1y} = \vec{r}_{2y}$, the x -components cancel and:

$$|\vec{\mu}_{\text{water}}| = \delta e \cdot 2r \cos\left(\frac{\theta}{2}\right), \quad \frac{\theta}{2} = 52.25, \quad \cos(52.25) = 0.612$$

Applying the O–H bond dipole moment equation $\mu_{\text{O-H}} = \delta e \cdot r$:

$$|\vec{\mu}_{\text{water}}| = 2 \mu_{\text{O-H}} \cos\left(\frac{\theta}{2}\right) = 2(1.5 \text{ D})(0.612) = 3.0 \text{ D} \times 0.612 \approx 1.836 \text{ D} \approx 1.8 \text{ D}$$

(Rounded to 2 significant figures, matching the O–H dipole value which has 2 sig figs.) The handwritten computation is shown in Figures ??.

Conclusion

Though the Wikipedia value implies suspiciously high precision, it aligns with experimental results. Since Clough et al. reported a five-significant-figure measurement, the Wikipedia value does not necessarily grossly misrepresent the precision achievable. However, we recommend caution: most scientific literature uses only 1.85 D (3 sig figs).

5.2. Modeling a Water Molecule as a Dipole

We model water as two point charges separated by a distance d , using the geometry from Section 5.1. We need the partial charge on each hydrogen and the effective charge-separation distance.

Partial charge

Let $q = \delta e$ be the partial charge on each hydrogen. From $|\vec{\mu}_{\text{water}}| = 2q r_{\text{OH}} \cos(\theta/2)$, solving for q :

$$q = \frac{|\mu_{\text{water}}|}{2 r_{\text{OH}} \cos(\theta/2)} = \frac{6.1709 \times 10^{-30} \text{ C}}{2(0.958 \times 10^{-10} \text{ m}) \cos(52.25)} \approx 5.26 \times 10^{-20} \text{ C}$$

$$\frac{q}{e} = \frac{5.26 \times 10^{-20}}{1.602 \times 10^{-19}} \approx 0.328e \approx 0.33e$$

Since the water molecule is neutral: $2q_H + q_O = 0 \Rightarrow q_O = -2q_H \approx -0.66e$.

The handwritten derivations are shown in Figures ??-??.

Effective dipole separation distance

Using the sine rule and Pythagorean theorem on the VSEPR geometry:

$$x = r \sin\left(\frac{\theta}{2}\right), \quad d = \sqrt{r^2 - r^2 \sin^2\left(\frac{\theta}{2}\right)} = r \sqrt{1 - \sin^2\left(\frac{\theta}{2}\right)} = r \cos\left(\frac{\theta}{2}\right)$$

So we can model a water molecule as two oppositely charged particles: $+0.33e$ and $-0.66e$, separated by $d = 0.587 \text{ \AA}$, as summarized in Figure ??.

Electric field between neighboring water molecules

According to Harvard's BioNumbers database, the average distance between water molecules in liquid state is $0.32 \text{ nm} = 3.2 \text{ \AA}$. We model a probe water molecule at this distance from our source dipole, positioned at equal distance y from the two charges. From the geometry:

$$3.2^2 = y^2 + \left(\frac{d}{2}\right)^2 \implies y = \sqrt{3.2^2 - \left(\frac{0.587}{2}\right)^2} \approx 3.24 \text{ \AA}$$

The handwritten field calculation is shown in Figure ??.

Our estimate for the field magnitude between two neighboring water molecules is:

$$E \approx 4.44 \times 10^{-9} \text{ V/m}$$

5.3. Electric Potential Energy Between Two Water Molecules

So far, our models did not account for the relative orientations of interacting dipoles. We use the formula from LibreTexts Chemistry for dipole–dipole potential energy:

$$U(r) = -\frac{p_1 p_2 k}{r_{12}^3} (\cos \theta_{12} - 3 \cos \theta_1 \cos \theta_2)$$

where p_1, p_2 are dipole moments, r_{12} is the center-to-center distance, θ_{12} is the angle between the two dipole moment vectors, and θ_1, θ_2 are the angles each dipole makes with the line connecting the two centers (Bertrand et al.).

Since we are dealing with identical molecules ($p_1 = p_2 = p$), the formula simplifies to:

$$U = -\frac{p^2 k}{r_{12}^3} (\cos \theta_{12} - 3 \cos \theta_1 \cos \theta_2)$$

We further factor as $U = S(r) \times F$, where the scaling factor $S(r) = p^2 k / r_{12}^3$ and $F = 3 \cos \theta_1 \cos \theta_2 - \cos \theta_{12}$.

The scaling factors for the three physical states of water, using $p = 1.85 \text{ D}$ and $k = 8.99 \times$

$10^9 \text{ N m}^2/\text{C}^2$:

$$S(0.31 \text{ nm}) = \frac{(1.85)^2 \times 8.99 \times 10^9}{(3.1 \times 10^{-10})^3} = 1.15 \times 10^{-20} \text{ J} = 0.0718 \text{ eV} \quad (\text{liquid})$$

$$S(0.28 \text{ nm}) = \frac{(1.85)^2 \times 8.99 \times 10^9}{(2.8 \times 10^{-10})^3} = 1.55 \times 10^{-20} \text{ J} = 0.0968 \text{ eV} \quad (\text{ice})$$

$$S(3.1 \text{ nm}) = \frac{(1.85)^2 \times 8.99 \times 10^9}{(3.1 \times 10^{-9})^3} = 1.15 \times 10^{-23} \text{ J} = 7.18 \times 10^{-5} \text{ eV} \quad (\text{vapor})$$

These values confirm that potential energy decreases as distance increases, due to the inverse cubic relationship. Distance does not affect the sign of U .

For four orientation configurations that yield integer values of F :

1. **Head to tail** (Figure ??): $\theta_1 = \theta_2 = \theta_{12} = 0$

$$F = 3 \cos 0 \cos 0 - \cos 0 = 3 - 1 = 2$$

$$U_{\text{liquid}} = 2 \times 0.0718 = 0.144 \text{ eV}$$

$$U_{\text{ice}} = 2 \times 0.0968 = 0.194 \text{ eV}$$

$$U_{\text{vapor}} = 2 \times 7.18 \times 10^{-5} = 1.44 \times 10^{-4} \text{ eV}$$

2. **Head to head** (Figure ??): $\theta_1 = 0$; $\theta_2 = 180$; $\theta_{12} = 0$

$$F = 3 \cos 0 \cos 180 - \cos 0 = -3 + 1 = -2$$

$$U_{\text{liquid}} = -2 \times 0.0718 = -0.144 \text{ eV}$$

$$U_{\text{ice}} = -2 \times 0.0968 = -0.194 \text{ eV}$$

$$U_{\text{vapor}} = -2 \times 7.18 \times 10^{-5} = -1.44 \times 10^{-4} \text{ eV}$$

3. **Side by side, same direction** (Figure ??): $\theta_1 = \theta_2 = 90$; $\theta_{12} = 0$

$$F = 3 \cos 90 \cos 90 - \cos 0 = 0 - 1 = -1$$

$$U_{\text{liquid}} = -1 \times 0.0718 = -0.0718 \text{ eV}$$

$$U_{\text{ice}} = -1 \times 0.0968 = -0.0968 \text{ eV}$$

$$U_{\text{vapor}} = -1 \times 7.18 \times 10^{-5} = -7.18 \times 10^{-5} \text{ eV}$$

4. **Side by side, opposite directions** (Figure ??): $\theta_1 = \theta_2 = 90$; $\theta_{12} = 180$

$$F = 3 \cos 90 \cos 90 - \cos 180 = 0 - (-1) = 1$$

$$U_{\text{liquid}} = 1 \times 0.0718 = 0.0718 \text{ eV}$$

$$U_{\text{ice}} = 1 \times 0.0968 = 0.0968 \text{ eV}$$

$$U_{\text{vapor}} = 1 \times 7.18 \times 10^{-5} = 7.18 \times 10^{-5} \text{ eV}$$

These results show that orientation affects not only the magnitude but also the sign of U . The head-to-tail configuration doubles the magnitude compared to the side-by-side case, while the side-by-side same-direction configuration changes the sign.

How Can Objects Have Solid Boundaries?

The electric field of a dipole falls off as $E \propto 1/x^3$, compared to a single charge's $1/x^2$. To understand why macroscopic objects appear to have sharp, well-defined edges, we examine what happens as we stack higher levels of charge symmetry.

A **quadrupole** can be thought of as two identical dipoles separated by a small distance d , pointing in opposite directions. The total potential is:

$$V_{\text{quad}}(x) = V_{+p}\left(x - \frac{d}{2}\right) + V_{-p}\left(x + \frac{d}{2}\right)$$

Using the dipole result $V_{\text{dipole}}(x) = kp/x^2$ and shifting:

$$V_{+p}(x - \frac{d}{2}) = \frac{k(+p)}{(x - \frac{d}{2})^2}, \quad V_{-p}(x + \frac{d}{2}) = \frac{k(-p)}{(x + \frac{d}{2})^2}$$

So:

$$V_{\text{quad}}(x) = kp \left(\frac{1}{(x - \frac{d}{2})^2} - \frac{1}{(x + \frac{d}{2})^2} \right)$$

For $x \gg d$, applying the binomial approximation $(x \pm a)^{-2} \approx x^{-2}(1 \mp 2a/x)$:

$$\frac{1}{(x - \frac{d}{2})^2} \approx \frac{1}{x^2} \left(1 + \frac{d}{x} \right), \quad \frac{1}{(x + \frac{d}{2})^2} \approx \frac{1}{x^2} \left(1 - \frac{d}{x} \right)$$

Subtracting:

$$V_{\text{quad}}(x) \approx kp \cdot \frac{1}{x^2} \cdot \frac{2d}{x} = \frac{2kpd}{x^3}$$

So $V_{\text{quad}} \propto 1/x^3$, falling off faster than the dipole ($1/x^2$). The electric field of the quadrupole is:

$$E_{x,\text{quad}} = -\frac{d}{dx} \left(\frac{2kpd}{x^3} \right) = \frac{6kpd}{x^4}$$

which falls off as $1/x^4$ — much faster than the dipole's $1/x^3$.

Putting it all together: as charge distributions become more symmetric (monopole $\rightarrow$ dipole $\rightarrow$ quadrupole $\rightarrow \dots$), the external field decays more rapidly:

$$E_{\text{monopole}} \propto \frac{1}{x^2}, \quad E_{\text{dipole}} \propto \frac{1}{x^3}, \quad E_{\text{quad}} \propto \frac{1}{x^4}, \quad \dots$$

This rapid decay means the influence of one atom's charges disappears just a few atomic distances away. In macroscopic objects, all these higher-order charge distributions overlap and cancel, so outside the material the net field is nearly zero. That creates a defined “edge” in how electric forces behave: inside, the fields are strong and complex; outside, they vanish. What we perceive as a solid edge is really the point where internal electric interactions stop having any effect beyond the material itself.

Deriving the Dipole–Dipole Force

7a. Force of dipole a on dipole b

Figure ?? shows the geometry of two aligned dipoles separated by center-to-center distance r .

From Section 2, the electric field of dipole a along its axis is $\vec{E}_x = 2kp_a/r^3$. The force of dipole a on dipole b is:

$$\vec{F}_{a \text{ on } b} = -q_b \vec{E}_a|_{r=r_-} + q_b \vec{E}_a|_{r=r_+} = -q_b \frac{2kp_a}{r_-^3} + q_b \frac{2kp_a}{r_+^3}$$

With $r_- = r - d_b/2$ and $r_+ = r + d_b/2$:

$$\vec{F}_{a \text{ on } b} = 2kp_a q_b \left[\left(r + \frac{d_b}{2} \right)^{-3} - \left(r - \frac{d_b}{2} \right)^{-3} \right]$$

Applying the binomial expansion $(a+x)^n \approx a^n + na^{n-1}x + \dots$ with $n = -3$, $a = r$, $x = \pm d_b/2$:

$$\left(r + \frac{d_b}{2} \right)^{-3} \approx r^{-3} - 3r^{-4} \left(\frac{d_b}{2} \right), \quad \left(r - \frac{d_b}{2} \right)^{-3} \approx r^{-3} + 3r^{-4} \left(\frac{d_b}{2} \right)$$

Substituting:

$$\boxed{F_{a \text{ on } b} = \frac{6k_e p_a p_b}{r^4}}$$

7b. Force on a polarizable molecule

If dipole b is a polarizable molecule with polarizability coefficient α_b , then its induced dipole moment is $p_b = \alpha_b E$, where $E = 2kp_a/r^3$ is the field from dipole a . Substituting:

$$p_b = \alpha_b \cdot \frac{2Kp_a}{r^3}$$

Plugging into $F_{a \text{ on } b} = 6k_e p_a p_b / r^4$:

$$F_{a \text{ on } b} = 6k_e \frac{p_a}{r^4} \left(\alpha_b \frac{2Kp_a}{r^3} \right) = \frac{12 k_e \alpha_b K p_a^2}{r^7}$$

This confirms that for a dipole interacting with a polarizable molecule, the force falls off as $1/r^7$.

Relevance of Van der Waals Forces

Van der Waals forces allow proteins to maintain their structure and shape when interacting with other proteins and materials, stabilizing the overall protein through weak attractive forces. Based on research by Charles M. Roth and colleagues, Van der Waals forces “decay as the inverse sixth power of separation distance,” meaning the force falls off as r^6 . This differs from our derivation, which gives r^7 . The discrepancy could be due to limited research on the topic at the time of publication. Despite this difference, our work shares several features with Roth’s: approximating the protein/dipole as a sphere and defining r as the center-to-center distance.

Van der Waals forces also appear in everyday objects. Microfiber cleaning cloths are made of polyester and nylon fibers measuring between 0.33 and $5\mu\text{m}$ across (Kaivac). These tiny fibers attract and hold dust, dirt, and other particles via Van der Waals forces. Each fiber exerts a small force, but the millions of fibers in a cloth combine to create a powerful cleaning tool. This “adhesive” behavior mimics the millions of micro-hairs on gecko feet, and the same mechanism makes microfiber cloths effective as antibacterial and antimicrobial cleaning resources (Berkshire).

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